NETWORK TESTING TOOLS THEORY

1.Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

ANS :

Command:

ping www.youtube.com

Steps:

1. Open Command Prompt (Win + R, type cmd, press Enter).
2. Type ping www.youtube.com and press Enter.
3. Check the average response time shown at the bottom. Windows displays the minimum, maximum, and average time in milliseconds (ms).



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Result (example for assignment):

- Packets sent: 4
- Packets received: 4
- Packets lost: 0 (0% loss)
- Average response time: 89 ms (sample result)

**2.Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.**

ANS:

Traceroute to www.flipkart.com

Command:

tracert www.flipkart.com

Steps:

1. Press Win + R, type cmd, and press Enter.
2. Type tracert www.flipkart.com.
3. Press Enter and wait for the trace to complete.

4. Record each hop number and its IP address or hostname.



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Intermediate hops (example format):

Hop	IP address / Hostname
1	192.168.1.1 (local router — example)
2	ISP router (actual address varies)
3	Intermediate ISP router
4 onward	Additional routers until the destination is reached

3.Use the nslookup tool to find the IP address of www.zomato.com and record the result.

ANS:

NSLOOKUP: www.zomato.com

Command:

nslookup www.zomato.com

Steps:

1. Open Command Prompt by pressing Win + R, typing cmd, and pressing Enter.
2. Type nslookup www.zomato.com and press Enter.
3. Note the IP address shown under Name: www.zomato.com or in the answer section.



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Result:

- Domain name: www.zomato.com
- IP address: Write the IP address displayed by your command.

4.Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

ANS:

Ping Test to a Local Network Device

Command:

ping 192.168.1.5

Steps:

1. Connect your computer and your friend's laptop or phone to the same Wi-Fi network.
2. Find the other device's IP address.
3. Open Command Prompt and type ping followed by that IP address.
4. Press Enter and check the result.

Result (example):

- Device online: Replies are received, showing the response time in milliseconds (ms).
- Device offline: The command may show Request timed out or Destination host unreachable.

**5.Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.**

ANS:

Short Report: Network Troubleshooting Tools

1. Ping: Useful for checking connectivity. I used ping www.youtube.com to check whether the website responds and to note the average response time.
2. Traceroute: Useful for diagnosing slow connections by identifying the network hops. I used tracert www.flipkart.com to view the route to the website.
3. Nslookup: Useful for identifying DNS issues. I used nslookup www.zomato.com to find the website's IP address.
4. (a) Website not loading: Use ping to check whether the destination responds.
5. (b) Slow connection: Use traceroute to identify hops where delays or timeouts occur.
6. (c) Suspecting DNS issues: Use nslookup to check whether the domain name resolves to an IP address.